VEL
SOFTWARE SOLUTIONS

HANDS-ON TRAINING

Practical Exercise Workbook

First manual-testing assignment plus eight practical employee projects with briefs, evidence checklists, acceptance criteria and marking rubrics

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1. First Manual-Testing Assignment (A0)

Test the **VEL Practice Shop** registration and login flow against a written specification, produce a traceable test suite, execute it, log defects and report results. Time-box: **4 working days**.

> **WHY THIS ASSIGNMENT FIRST**
>
> It exercises the whole manual-testing chain in miniature: read a requirement, design cases, run them, capture evidence, report a defect, and summarise status. Every later project repeats this chain with more scale or more tooling.

Specification under test

ID	Requirement
REQ-001	Registration: **Full name** is required, 2–40 characters after trimming spaces.
REQ-002	Registration: **Email** is required and must be a valid address (`name@domain.tld`).
REQ-003	Registration: **Age** is required, a whole number from **18 to 60 inclusive**.
REQ-004	Registration: **Password** 8–20 characters with at least one digit and at least one uppercase letter.
REQ-005	Registration: **Confirm password** must match the password exactly (case-sensitive).
REQ-006	Registration: the **Terms** checkbox must be ticked.
REQ-007	A successful registration shows “Registration successful” and opens the login view.
REQ-008	Login with registered credentials opens the shop. Any failure shows the generic message “Invalid email or password”.
REQ-009	On the **3rd consecutive failed login** the account locks for 30 seconds and the message “Account locked. Try again in 30 seconds.” is shown; attempts during the lock show the same message.
REQ-010	Shop: quantity per item is a whole number **1–10**. Cart total = sum(price × quantity) shown to 2 decimals; GST 18 % shown to 2 decimals (standard rounding).
REQ-011	Session expires after **90 seconds of inactivity**: “Session expired. Please log in again.” and the login view opens.
REQ-012	Logout ends the session. Using the browser Back button afterwards must **not** show shop content.
REQ-013	The application is usable at **360 px** viewport width with no horizontal scrolling.
REQ-014	Accessibility: every input has a programmatic label; every image has alt text; all functions work by keyboard with a visible focus indicator.

Scope

In scope	Out of scope
Registration view, login view, shop view (cart, quantity, totals), session timeout, logout, viewport 360 px and 1280 px, keyboard access.	Performance, security exploitation, other devices, real e-mail delivery, payment. Report any security concern you notice to your buddy instead of probing it.

Procedure

Day	Steps	Deliverable
1	1. Read REQ-001 to REQ-014 twice; list ambiguities and unanswered questions. 2. Run the app; explore for 30 minutes without a script and note what you observe (exploratory notes). 3. Create the requirements traceability matrix with one row per requirement.	`docs/requirements-traceability.md` with all 14 rows; `docs/questions.md` ; exploratory notes
2	1. For each requirement apply **equivalence partitioning** and **boundary-value analysis** where it has ranges (REQ-001, 003, 004, 010). 2. Write at least **30 test cases** using the template: ID, title, requirement, preconditions, steps, test data, expected result, priority. 3. Include positive and negative cases; at least 6 for each of REQ-003 and REQ-004.	`test-cases/functional/registration-login.csv` (or tool export), 30+ cases, each linked to a REQ
3	1. Execute every case on Chrome and Firefox at 1280 px; REQ-013 at 360 px. 2. Record Pass / Fail / Blocked with evidence file names per the convention. 3. Raise a defect (template) the moment you confirm a failure; attempt to reproduce it three times.	Filled execution log; `evidence/` files; `defects/DEF-nnn.md` for each defect
4	1. Re-check traceability: every requirement has at least one executed case. 2. Complete the test execution report with counts, defect list, coverage and your own risk statement. 3. Present for 10 minutes to the QA Lead.	`docs/test-execution-report.md` ; review notes from the QA Lead

Worked example: one test case and one defect

Use these as the quality bar. Notice that the expected result is **observable and unambiguous**, and the defect states exactly what happened versus what was required.

Test case ID	`TC-REG-004` · Requirement: REQ-003 · Technique: boundary-value analysis · Priority: High
Title	Registration accepts age 18 (lower boundary)
Preconditions	Practice Shop open at the Register view; no account exists for `bva18@example.test` .
Steps	1. Enter name `Asha Rao` . 2. Enter email `bva18@example.test` .

	3. Enter age `18`. 4. Enter password `Passw0rdOK` and the same value in Confirm password. 5. Tick Terms. 6. Click **Register**.
Expected result	The message “Registration successful” is displayed and the Login view opens. No error is shown next to Age.
Actual result / status	Fill in during execution: **PASS** / **FAIL** / **BLOCKED** + evidence file name.

Defect ID / title	`DEF-001` · Age 18 rejected although REQ-003 allows 18 to 60 inclusive
Severity / priority	Major (valid users cannot register) / High
Environment	Practice Shop 1.0, Chrome 130 on Windows 11, viewport 1280×800, 2026-11-06
Steps to reproduce	1. Open Register view. 2. Fill all fields validly, using age `18`. 3. Tick Terms and click **Register**.
Expected	Registration succeeds (18 is the inclusive lower limit).
Actual	Age field shows an error; registration is blocked. Reproduced 3 of 3 attempts.
Evidence	`evidence/2026-11-06_TC-REG-004_FAIL_1.png`

Acceptance criteria and marking (100 points)

Criterion	Points	Full marks when
Traceability	15	All 14 requirements appear in the matrix and each links to at least one executed case.
Test design quality	25	Equivalence classes and boundaries are identified and named; cases are atomic, repeatable and have measurable expected results.
Execution discipline	20	Every case has a status and evidence; results are on two browsers; REQ-013 checked at 360 px.
Defect reports	25	Each defect is reproducible by someone else from the report alone; severity and priority are justified; no duplicates.
Report and honesty	15	Counts reconcile (pass + fail + blocked + not run = total); risks and untested areas are stated plainly.

PASS MARK: 70 / 100

Below 70 the trainee repeats the weakest part with feedback. The QA Lead keeps a hidden answer key of seeded defects (never shown to trainees). Finding more defects does not earn extra marks; **report quality** does.

2. Practical Employee Projects

Eight projects take the trainee from manual testing to a complete client QA report. Each project is marked on the same rubric and must be reviewed by a person other than its author. Targets are the local practice systems or an explicitly authorised site.

#	Project	Level	Time	Primary deliverable
1	Test a business website manually	Level 3	5 days	`docs/test-plan.md` (lite)
2	Prepare a complete test case suite	Level 2	5 days	`test-cases/functional/shop.csv`
3	Test a REST API	Level 4	5 days	`automation/api/postman/collection.json`
4	Automate browser tests with Playwright	Level 5	5 days	`automation/ui/`
5	Automate API tests	Level 5	4 days	`automation/api/`
6	Create a regression suite	Level 2 + 5	3 days	`test-cases/regression/`
7	Perform a controlled performance test	Level 6	4 days	`performance/` scripts
8	Prepare a complete client QA report	All levels	5 days	Complete `qa-project/` repository

AUTHORISATION FIRST

Only test the Practice Shop and Practice API (running on your own machine) or a site whose owner has authorised testing in writing. Never run automation, load or scanning tools against public websites, third-party APIs or shared systems without that authorisation.

Project 1 — Test a business website manually

Level / time-box	Level 3 · 5 days
Target system	The company's own website, or a client site **with written authorisation**, or a public practice site. No authorisation, no testing.
Tasks	1. Write a one-page scope and test approach (what is in/out, browsers, devices, data). 2. Build a sitemap; list user journeys and risks. 3. Design and execute at least **40 test cases**: navigation, links, forms, content, responsive layout (360/768/1280 px), cross-browser (two browsers), first-line accessibility. 4. Log every defect with evidence; rate severity and priority. 5. Write the test execution report with an honest risk statement.
Deliverables (repository paths)	`docs/test-plan.md` (lite), `test-cases/functional/website.csv`, `evidence/`, `defects/`, `docs/test-execution-report.md`.

Acceptance criteria	Every case executed with evidence; ≥ 5 reproducible defects or documented zero-defect reasoning; counts reconcile; report states what was **not** tested.
Marking (100 points)	Planning and approach 15 · Technical quality 30 · Evidence and traceability 20 · Reporting honesty 20 · Professional conduct and timeliness 15. **Pass: 70.**

Project 2 — Prepare a complete test case suite

Level / time-box	Level 2 · 5 days
Target system	VEL Practice Shop, shop/cart/session/logout modules (REQ-010 to REQ-014) plus registration regression.
Tasks	1. Create the requirements traceability matrix. 2. Apply EP, BVA, decision table, state transition and error guessing; name the technique on each case. 3. Write at least **60 test cases** across functional, negative, boundary, session and accessibility types, each linked to a requirement. 4. Tag cases: smoke, regression, UAT, automation candidate. 5. Arrange a four-eyes review with a colleague; record and resolve every comment.
Deliverables (repository paths)	`test-cases/functional/shop.csv`, `docs/requirements-traceability.md`, review log.
Acceptance criteria	100 % requirement coverage; no duplicate cases; ≥ 25 % of cases are negative or boundary; review comments closed.
Marking (100 points)	Planning and approach 15 · Technical quality 30 · Evidence and traceability 20 · Reporting honesty 20 · Professional conduct and timeliness 15. **Pass: 70.**

Project 3 — Test a REST API

Level / time-box	Level 4 · 5 days
Target system	VEL Practice API (`qa-project-template/practice-api`) running locally.
Tasks	1. Read the contract and build a Postman collection with ≥ **40 requests**. 2. Cover positive, negative, authentication, boundary, schema and method tests for every endpoint. 3. Add assertions on status, headers, body, schema and response time. 4. Raise a defect for each deviation, with request/response evidence (tokens removed). 5. Hand over any access-control observation as a **security escalation** note, not a public defect.
Deliverables (repository paths)	`automation/api/postman/collection.json`, defects, `docs/api-test-report.md`.
Acceptance criteria	≥ 8 correct defects; no false defects against a documented behaviour; all endpoints covered; escalation note present.

Marking (100 points)	Planning and approach 15 · Technical quality 30 · Evidence and traceability 20 · Reporting honesty 20 · Professional conduct and timeliness 15. **Pass: 70.**

Project 4 — Automate browser tests with Playwright

Level / time-box	Level 5 · 5 days
Target system	VEL Practice Shop UI.
Tasks	1. Set up Playwright with pytest; configure base URL through an environment variable. 2. Create page objects for Register, Login and Shop. 3. Automate ≥ **15 tests** including parametrised boundary tests; mark known defects `xfail(strict=True)` with the defect ID. 4. Run on Chromium and Firefox; enable trace on first retry/failure. 5. Run each test 5 times; fix or remove any flaky test; document how to run.
Deliverables (repository paths)	`automation/ui/`, `automation/page-objects/`, `automation/fixtures/`, HTML/JUnit report.
Acceptance criteria	Suite green on 5 consecutive runs; no sleeps; no secrets; code review passed.
Marking (100 points)	Planning and approach 15 · Technical quality 30 · Evidence and traceability 20 · Reporting honesty 20 · Professional conduct and timeliness 15. **Pass: 70.**

Project 5 — Automate API tests

Level / time-box	Level 5 · 4 days
Target system	VEL Practice API.
Tasks	1. Create session fixtures for base URL and token (credentials from environment variables). 2. Write ≥ **15 pytest tests**, including JSON-schema validation of products. 3. Mark seeded defects as strict `xfail` with defect IDs. 4. Generate JUnit XML; run in CI. 5. Report automation coverage against the Level-4 manual cases.
Deliverables (repository paths)	`automation/api/`, `automation/api/schemas/`, CI run link, coverage note.
Acceptance criteria	Independent tests; zero shared state; schema checks present; CI green.
Marking (100 points)	Planning and approach 15 · Technical quality 30 · Evidence and traceability 20 · Reporting honesty 20 · Professional conduct and timeliness 15. **Pass: 70.**

Project 6 — Create a regression suite

Level / time-box	Level 2 + 5 · 3 days

Target system	Practice Shop build 1.0 vs build 1.1 (`?build=1.1`).
Tasks	1. Classify tests into tiers: **smoke** (≤ 15 min), **regression** (nightly), **full** (weekly). 2. Write selection rules and a change-impact table. 3. Run the suite on build 1.0 (baseline) and build 1.1. 4. Compare results; identify fixed defects, **new regressions**, and obsolete tests. 5. Report with a go/no-go style recommendation for build 1.1.
Deliverables (repository paths)	`test-cases/regression/` , `docs/regression-report.md` , updated RTM.
Acceptance criteria	At least one regression found and explained; every skipped test justified; recommendation supported by evidence.
Marking (100 points)	Planning and approach 15 · Technical quality 30 · Evidence and traceability 20 · Reporting honesty 20 · Professional conduct and timeliness 15. **Pass: 70.**

Project 7 — Perform a controlled performance test

Level / time-box	Level 6 · 4 days
Target system	VEL Practice API on **localhost only** (never a shared or third-party system).
Tasks	1. Write a one-page performance test plan: objectives, workload model (use the worked formula), thresholds (e.g. p95 < 500 ms, errors < 1 %), environment, data, stop conditions. 2. Script `GET /api/products` and `GET /api/slow?delay=200` in k6, JMeter or Locust; validate with one user. 3. Run baseline (1 VU), then load at 10, 25 and 50 VUs with ramp-up, steady state and ramp-down. 4. Record client metrics and the server's CPU/memory (e.g. `top`). 5. Report p50/p95/p99, throughput, errors, bottleneck notes and **limitations** (local machine, synthetic data).
Deliverables (repository paths)	`performance/` scripts, results summary, `docs/performance-report.md` .
Acceptance criteria	Plan approved before running; steady-state only is measured; limitations stated; no claim beyond measured data.
Marking (100 points)	Planning and approach 15 · Technical quality 30 · Evidence and traceability 20 · Reporting honesty 20 · Professional conduct and timeliness 15. **Pass: 70.**

Project 8 — Prepare a complete client QA report

Level / time-box	All levels · 5 days
Target system	Run the full lifecycle on the Practice Shop + Practice API as if it were a paying client.
Tasks	1. Produce strategy, plan, RTM, test cases, execution report, defect summary, automation coverage report, performance and accessibility summaries.

	2. Compute all metrics with the formulas in this handbook. 3. Write the release QA report with the recommendation (Go / Conditional Go / No-Go) and the unresolved-risk summary. 4. Complete the QA sign-off checklist; the QA Lead acts as second reviewer. 5. Present for 20 minutes to a mock client panel.
Deliverables (repository paths)	Complete `qa-project/` repository, `docs/release-report.md`, `docs/qa-signoff.md`, presentation.
Acceptance criteria	Numbers reconcile with source data; recommendation is consistent with evidence; risks are plain; panel questions answered honestly.
Marking (100 points)	Planning and approach 15 · Technical quality 30 · Evidence and traceability 20 · Reporting honesty 20 · Professional conduct and timeliness 15. **Pass: 70.**

Project sign-off sheet

Project	Date	Score /100	Reviewer	Pass (≥ 70)	Reviewer comments
1. Test a business website manually				☐	
2. Prepare a complete test case suite				☐	
3. Test a REST API				☐	
4. Automate browser tests with Playwright				☐	
5. Automate API tests				☐	
6. Create a regression suite				☐	
7. Perform a controlled performance test				☐	
8. Prepare a complete client QA report				☐	